

Eraker, Johannes, and Polson (2003): The Impact of Jumps in Volatility and Returns

Introduction

A well-documented finding in the literature is that models combining diffusive stochastic volatility with jumps in returns — such as the SVJ model of Bates (1996) — are still misspecified. The core problem is that volatility in these models can only move gradually, driven by normally distributed diffusion shocks, yet empirical data shows that conditional volatility can increase very rapidly around periods of market stress. Eraker, Johannes, and Polson (2003) address this gap by introducing jumps directly into the volatility process, arguing that such a factor is rapidly moving, like jumps in returns, but also persistent, unlike them. The paper provides the first likelihood-based empirical evidence for the presence of volatility jumps using S&P 500 and Nasdaq 100 index return data.

The Model

The paper considers four nested models. Writing $Y_t = \log S_t$ for the log-price and V_t for the instantaneous variance, the most general dynamics are:

$$\begin{pmatrix} dY_t \\ dV_t \end{pmatrix} = \begin{pmatrix} \mu \\ \kappa(\theta - V_{t-}) \end{pmatrix} dt + \sqrt{V_{t-}} \begin{pmatrix} 1 & 0 \\ \rho \sigma_v & \sqrt{1 - \rho^2} \cdot \sigma_v \end{pmatrix} dW_t + \begin{pmatrix} \xi^y dN_t^y \\ \xi^v dN_t^v \end{pmatrix} \quad (1)$$

where W_t is a standard Brownian motion in $\mathbb{R}^2$, N_t^y and N_t^v are Poisson processes with intensities λ_y and λ_v , and ξ^y , ξ^v are the jump sizes in returns and volatility respectively. This nests four models: the pure stochastic volatility model **SV** (no jumps), the **SVJ** model (jumps in returns only, $\xi^v \sim N(\mu_y, \sigma_y^2)$, $\lambda_v = 0$), the **SVCJ** model (contemporaneous jumps in both returns and volatility, $N_t^y = N_t^v = N_t$, with $\xi^v \sim \text{Exp}(\mu_v)$ and $\xi^y | \xi^v \sim N(\mu_y + \rho_J \xi^v, \sigma_y^2)$), and the **SVIJ** model (independent jumps in returns and volatility). The exponential distribution for volatility jump sizes guarantees V_t stays positive and captures the large upward spikes in volatility observed empirically.

The intuition behind including volatility jumps is straightforward. Jumps in returns generate large discrete moves but have no lasting impact on future return distributions. Diffusive volatility is persistent but moves slowly. Volatility jumps combine both features: they arrive suddenly and cause an immediate, large increase in V_t , which then mean-reverts gradually back to its long-run level. This gives the model a mechanism to generate the kind of rapid, persistent volatility increases observed around crash episodes.

Estimation via MCMC

The paper estimates all four models using Markov Chain Monte Carlo (MCMC), a likelihood-based approach that simultaneously recovers parameters, latent spot volatilities, jump times, and jump sizes from the return time series. This goes beyond standard calibration methods, which only estimate parameters and treat volatility as unobserved. The MCMC posterior distribution is:

$$p(\Theta, J, \xi^y, \xi^v, V \mid Y) \propto p(Y \mid \Theta, J, \xi^y, \xi^v, V) p(\Theta, J, \xi^y, \xi^v, V) \quad (2)$$

where J , ξ^y , ξ^v , and V are vectors of jump times, jump sizes, and spot volatilities over the sample. The algorithm iteratively draws from the conditional posteriors of each block, producing samples that fully quantify parameter and volatility estimation uncertainty. Model comparison is carried out using log-Bayes factors, which penalise model complexity automatically and require no appeal to asymptotic distributions.

Empirical Results and Option Pricing Implications

Estimated on daily S&P 500 returns from 1980 to 1999, the SV model is severely misspecified: generating the -23% crash return of October 1987 requires an almost eight standard deviation diffusion shock. Adding jumps in returns (SVJ) improves the fit but introduces clustered jump arrivals during the crash week — three estimated jumps in a single week — which contradicts the i.i.d. Poisson assumption. Log-Bayes factors strongly favour the models with volatility jumps: the log-Bayes factor comparing SV to SVIJ is 59.45 for the S&P and 56.03 for the Nasdaq, constituting overwhelming evidence. In the SVCJ model, the 1987 crash is explained by a single return jump of -14% combined with a volatility jump that pushed V_t from 40% to over 50%, a much more economically coherent picture than the SVJ model’s cluster of jumps.

For option pricing, jumps in returns steepen the slope of the implied volatility curve, while volatility jumps additionally raise implied volatility for deep in-the-money options, producing the characteristic hook or “tipping at the end” effect documented by Duffie, Pan, and Singleton (2000). The SV model generates a nearly flat implied volatility curve because it produces negligible conditional non-normalities at short horizons. Estimation risk also matters: spot volatility uncertainty dominates option price uncertainty for short-dated at-the-money options, while parameter uncertainty dominates for out-of-the-money options and long maturities.

Strengths, Limitations, and Relevance to QQQ Options

The paper’s main contribution is providing direct empirical evidence, from return data alone, that both types of jumps are necessary components of a well-specified model. The MCMC approach is a genuine methodological advance because it recovers the full time series of latent states, making it possible to attribute specific market episodes to specific model components. The use of long samples spanning the 1987, 1997, and 1998 stress episodes is especially valuable for identifying rare jump events that shorter option-based studies miss.

The main limitations follow those of the simpler jump-diffusion models. Jump intensities λ_y and λ_v are kept constant, so the model cannot capture the time-varying crash risk that is visible in option markets. The constant intensity assumption is directly contradicted by the paper’s own finding that the jump component varies strongly over time, echoing the conclusion of Carr and Wu (2003). Calibration using only return data also means that risk premia embedded in option prices are not directly estimated, which limits the model’s immediate use for option pricing without additional assumptions.

For our QQQ project, the Eraker et al. framework provides important context. The Nasdaq 100 results in the paper show that Nasdaq volatility is systematically higher and more volatile than S&P volatility, with jumps arriving roughly three times more frequently. This is consistent with the technology-heavy composition of the QQQ and reinforces the need for a model that allows both return and volatility jumps. The finding that jumps in volatility, not diffusive stochastic volatility, are the primary drivers of market stress episodes suggests that option pricing models applied to QQQ should allow for stochastic and jump-driven volatility to avoid the systematic mispricing of short-dated out-of-the-money puts that simpler models produce.